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https://www.tensorflow.org/api_docs/python/tf/scatter_nd

Scatters updates into a tensor of shape shape according to indices.

Code Examples

```
tf.scatter_nd(  
    indices: Annotated[Any, TV_ScatterNd_Tindices],  
    updates: Annotated[Any, TV_ScatterNd_T],  
    shape: Annotated[Any, TV_ScatterNd_Tindices],  
    name=None  
) -> Annotated[Any, TV_ScatterNd_T]
```

```
tf.scatter_nd(  
    indices: Annotated[Any, TV_ScatterNd_Tindices],  
    updates: Annotated[Any, TV_ScatterNd_T],  
    shape: Annotated[Any, TV_ScatterNd_Tindices],  
    name=None  
) -> Annotated[Any, TV_ScatterNd_T]
```

```
indices.shape[-1] <= shape.rank
```

```
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```

```
indices.shape[:-1] + shape[indices.shape[-1]:]
```

```
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```

```
indices = tf.constant([[4], [3], [1], [7]])  
updates = tf.constant([9, 10, 11, 12])  
shape = tf.constant([8])  
scatter = tf.scatter_nd(indices, updates, shape)  
print(scatter)
```

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indices = tf.constant([[4], [3], [1], [7]])  
updates = tf.constant([9, 10, 11, 12])  
shape = tf.constant([8])  
scatter = tf.scatter_nd(indices, updates, shape)  
print(scatter)
```

Documentation

Scatters updates into a tensor of shape shape according to indices.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.manip.scatter_nd`, `tf.compat.v1.scatter_nd`

Used in the notebooks

- Introduction to tensor slicing

Scatter sparse updates according to individual values at the specified indices. This op returns an output tensor with the shape you specify. This op is the inverse of the `tf.gather_nd` operator which extracts values or slices from a given tensor.

This operation is similar to `tf.tensor_scatter_nd_add`, except that the tensor is zero-initialized. Calling `tf.scatter_nd(indices, updates, shape)` is identical to calling `tf.tensor_scatter_nd_add(tf.zeros(shape, updates.dtype), indices, updates)`

If indices contains duplicates, the associated updates are accumulated (summed) into the output tensor.

`indices` is an integer tensor containing indices into the output tensor. The last dimension of `indices` can be at most the rank of `shape`:

The last dimension of `indices` corresponds to indices of elements (if `indices.shape[-1] = shape.rank`) or slices (if `indices.shape[-1] < shape.rank`) along dimension `indices.shape[-1]` of `shape`.

`updates` is a tensor with shape:

The simplest form of the scatter op is to insert individual elements in a tensor by index. Consider an example where you want to insert 4 scattered elements in a rank-1 tensor with 8 elements.

In Python, this scatter operation would look like this:

The resulting tensor would look like this:

You can also insert entire slices of a higher rank tensor all at once. For example, you can insert two slices in the first dimension of a rank-3 tensor with two matrices of new values.

In Python, this scatter operation would look like this:

The resulting tensor would look like this:

Note that on CPU, if an out of bound index is found, an error is returned. On GPU, if an out of bound index is found, the index is ignored.

Returns